

GEN211 Probability and Statistics

Lecture 4: An Introduction to Correlation, Interpolation, and Curve Fitting





Lecture contents

- An introduction to correlation
- An introduction to interpolation
- An introduction to curve fitting



An Introduction to Correlation



Inferential statistics from another perspective

- In addition to confidence intervals and hypothesis testing, another area of inferential statistics involves determining whether a relationship exists between two or more numerical or quantitative variables.
- For instance, a businessperson may want to know whether the volume of sales for a given month is related to the amount of advertising the firm does that month.
- Also, educators need to determine whether the number of hours a student studies is related to the student's score on a particular exam.
- Another example, medical researchers are interested in questions such as, is caffeine related to heart damage? or is there a relationship between a person's age and his or her blood pressure?
- These are only some of the many questions that can be answered using the techniques of correlation and regression analysis.



Scatter plot

In simple correlation and regression studies, the researcher collects data on two numerical or quantitative variables to see whether a relationship exists between the variables. For example, if a researcher wishes to see whether there is a relationship between number of hours of study and test scores on an exam, she must select a random sample of students, determine the number of hours each studied, and obtain their grades on the exam. A table can be made for the data, as shown here.

Student	Hours of study x	Grade y (%)
A	6	82
B	2	63
C	1	57
D	5	88
E	2	68
F	3	75

The two variables for this study are called the **independent variable** and the **dependent variable**.



Scatter plot

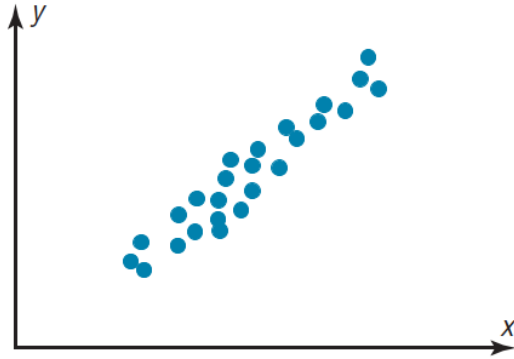
The independent and dependent variables can be plotted on a graph called a *scatter plot*. The independent variable x is plotted on the horizontal axis, and the dependent variable y is plotted on the vertical axis.

A **scatter plot** is a graph of the ordered pairs (x, y) of numbers consisting of the independent variable x and the dependent variable y .

The scatter plot is a visual way to describe the nature of the relationship between the independent and dependent variables. The scales of the variables can be different, and the coordinates of the axes are determined by the smallest and largest data values of the variables.



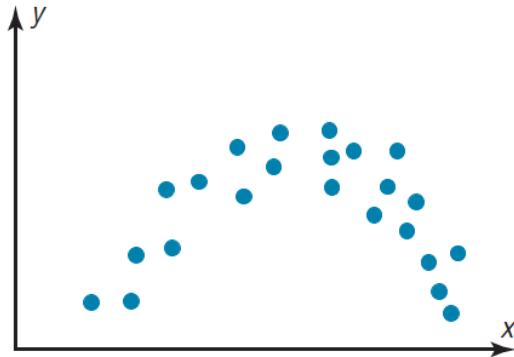
Scatter plot



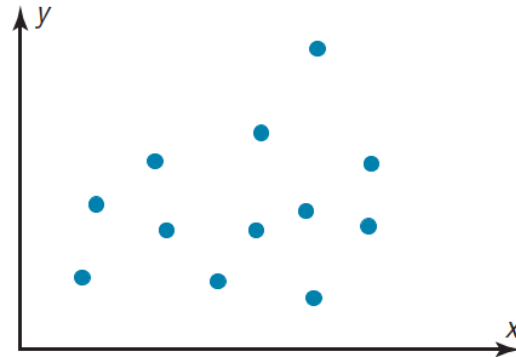
(a) Positive linear relationship



(b) Negative linear relationship



(c) Curvilinear relationship



(d) No relationship



Scatter plot

Procedure Table

Drawing a Scatter Plot

- | | |
|---------------|--|
| Step 1 | Draw and label the x and y axes. |
| Step 2 | Plot each point on the graph. |
| Step 3 | Determine the type of relationship (if any) that exists for the variables. |



Example 1

Construct a scatter plot for the data shown for car rental companies in the United States for a recent year.

Company	Cars (in ten thousands)	Revenue (in billions)
A	63.0	\$7.0
B	29.0	3.9
C	20.8	2.1
D	19.1	2.8
E	13.4	1.4
F	8.5	1.5

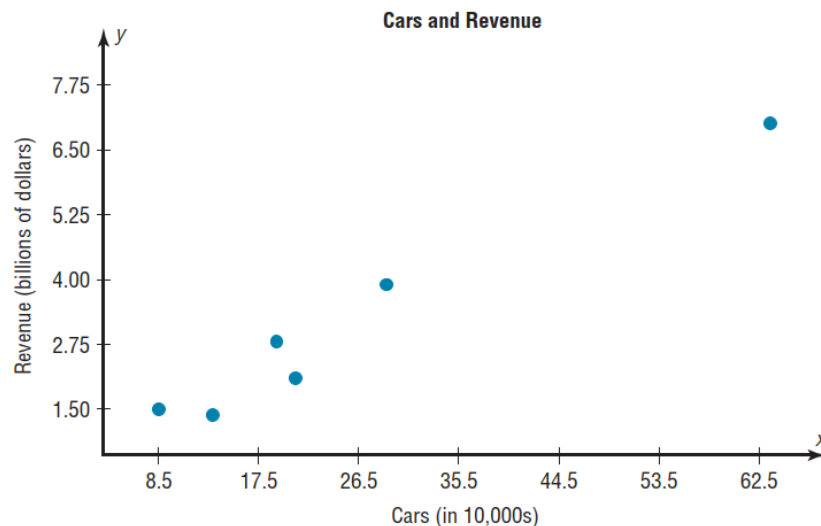


Example 1

SOLUTION

Step 1 Draw and label the x and y axes.

Step 2 Plot each point on the graph, as shown.



Step 3 Determine the type of relationship (if any) that exists.

In this example, it looks as if a positive linear relationship exists between the number of cars that an agency owns and the total revenue that is made by the company.



Example 2

Construct a scatter plot for the data obtained in a study on the number of absences and the final grades of seven randomly selected students from a statistics class. The data are shown here.

Student	Number of absences x	Final grade y (%)
A	6	82
B	2	86
C	15	43
D	9	74
E	12	58
F	5	90
G	8	78

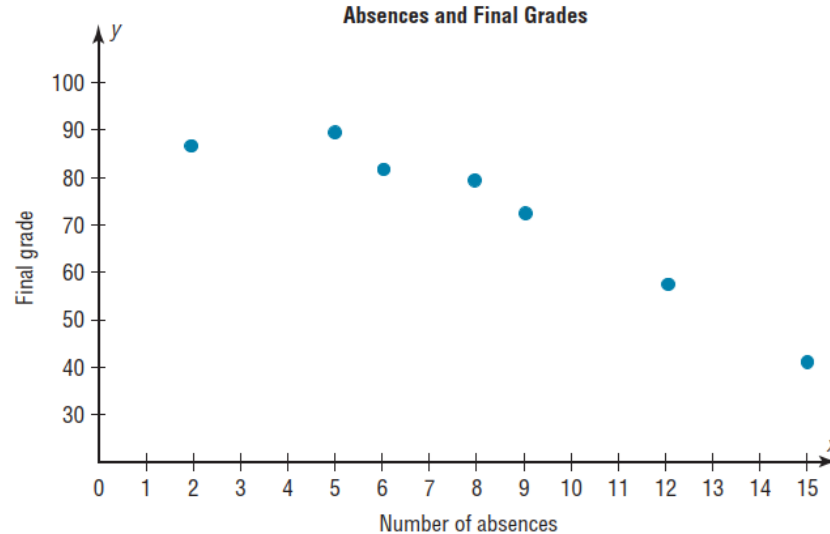


Example 2

SOLUTION

Step 1 Draw and label the x and y axes.

Step 2 Plot each point on the graph, as shown.



Step 3 Determine the type of relationship (if any) that exists.

In this example, it looks as if a negative linear relationship exists between the number of student absences and the final grade of the students.



Example 3

A researcher wishes to see if there is a relationship between the ages of the wealthiest people in the world and their net worth. A random sample of 10 persons was selected from the *Forbes* list of the 400 richest people for a recent year. The data are shown. Draw a scatter plot for the data.

Person	Age x	Net worth y (in billions of dollars)
A	60	11
B	72	69
C	56	11.9
D	55	30
E	83	12.2
F	67	36
G	38	18.7
H	62	10.2
I	62	23.3
J	46	10.6

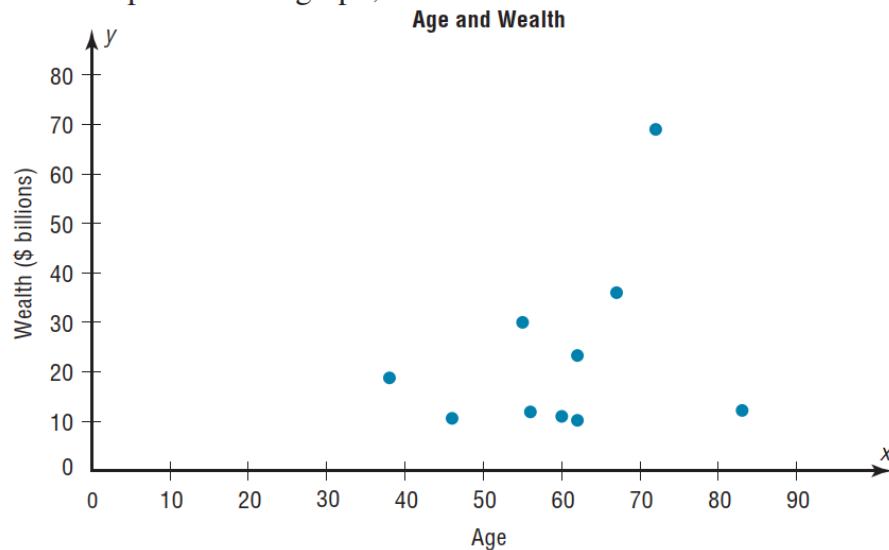


Example 3

SOLUTION

Step 1 Draw and label the x and y axes.

Step 2 Plot each point on the graph, as shown.



Step 3 Determine the type of relationship (if any) that exists.

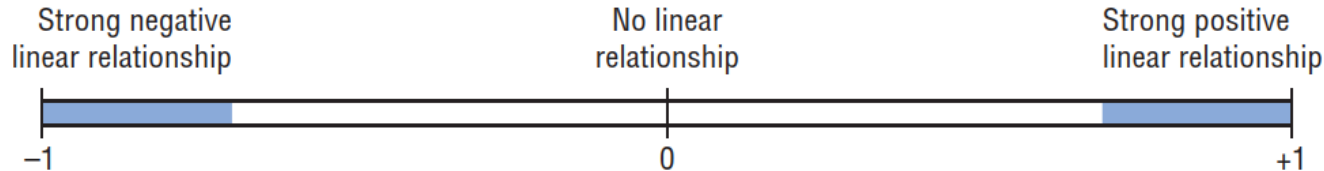
In this example, there is no type of a strong linear or curvilinear relationship between a person's age and his or her net worth.



Correlation

The **linear correlation coefficient** computed from the sample data and measures the strength and direction of a linear relationship between two quantitative variables. The symbol for the sample correlation coefficient is r .

The *range of the linear correlation coefficient* is from -1 to $+1$. If there is a *strong positive linear relationship* between the variables, the value of r will be close to $+1$. If there is a *strong negative linear relationship* between the variables, the value of r will be close to -1 . When there is no linear relationship between the variables or only a weak relationship, the value of r will be close to 0 . When the value of r is 0 or close to zero, it implies only that there is no linear relationship between the variables. The data may be related in some other nonlinear way.



Correlation

Properties of the Linear Correlation Coefficient

1. The correlation coefficient is a unitless measure.
2. The value of r will always be between -1 and $+1$ inclusively. That is, $-1 \leq r \leq 1$.
3. If the values of x and y are interchanged, the value of r will be unchanged.
4. If the values of x and/or y are converted to a different scale, the value of r will be unchanged.
5. The value of r is sensitive to outliers and can change dramatically if they are present in the data.



Correlation

Assumptions for the Correlation Coefficient

1. The sample is a random sample.
2. The data pairs fall approximately on a straight line and are measured at the interval or ratio level.
3. The variables have a bivariate normal distribution. (This means that given any specific value of x , the y values are normally distributed; and given any specific value of y , the x values are normally distributed.)

Formula for the Linear Correlation Coefficient r

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n(\sum x^2) - (\sum x)^2][n(\sum y^2) - (\sum y)^2]}}$$

where n is the number of data pairs.



Correlation

Procedure Table

Finding the Value of the Linear Correlation Coefficient

Step 1 Make a table as shown.

x	y	xy	x^2	y^2
-----	-----	------	-------	-------

Step 2 Place the values of x in the x column and the values of y in the y column. Multiply each x value by the corresponding y value, and place the products in the xy column.

Square each x value and place the squares in the x^2 column.

Square each y value and place the squares in the y^2 column.

Find the sum of each column.

Step 3 Substitute in the formula and find the value for r .

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n(\sum x^2) - (\sum x)^2][n(\sum y^2) - (\sum y)^2]}}$$

where n is the number of data pairs.



Example 4, revisit slide 11

Compute the value of the linear correlation coefficient for the data obtained in the study of the number of absences and the final grade of the seven students in the statistics class given in Example 2.

SOLUTION

Step 1 Make a table.

Step 2 Find the values of xy , x^2 , and y^2 ; place these values in the corresponding columns of the table.

Student	Number of absences x	Final grade y (%)	xy	x^2	y^2
A	6	82	492	36	6,724
B	2	86	172	4	7,396
C	15	43	645	225	1,849
D	9	74	666	81	5,476
E	12	58	696	144	3,364
F	5	90	450	25	8,100
G	8	78	624	64	6,084
	$\Sigma x = 57$	$\Sigma y = 511$	$\Sigma xy = 3745$	$\Sigma x^2 = 579$	$\Sigma y^2 = 38,993$



Example 4

SOLUTION

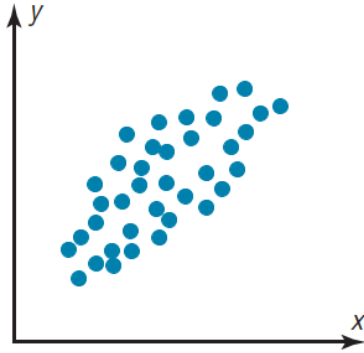
Step 3 Substitute in the formula and solve for r :

$$\begin{aligned} r &= \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n(\sum x^2) - (\sum x)^2][n(\sum y^2) - (\sum y)^2]}} \\ &= \frac{(7)(3745) - (57)(511)}{\sqrt{[(7)(579) - (57)^2][(7)(38,993) - (511)^2]}} = -0.944 \end{aligned}$$

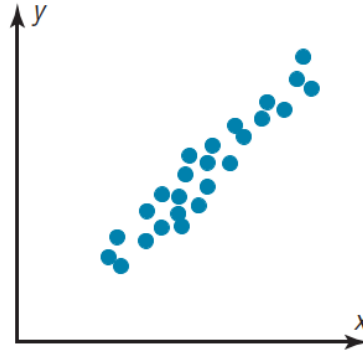
The value of r suggests a strong negative linear relationship between a student's final grade and the number of absences a student has. That is, the more absences a student has, the lower is his or her grade.



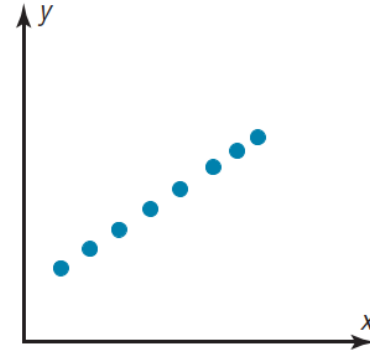
Linear correlation coefficient vs. scatter plot



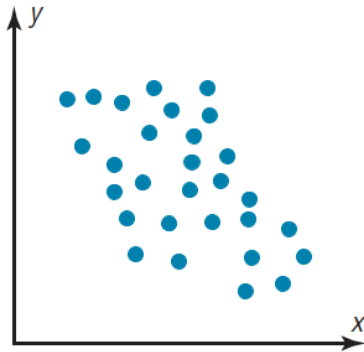
(a) $r = 0.50$



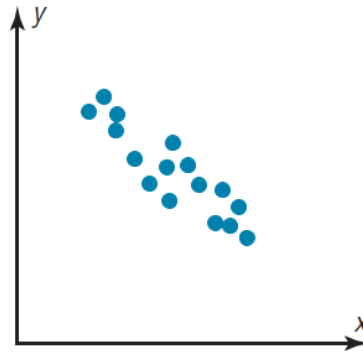
(b) $r = 0.90$



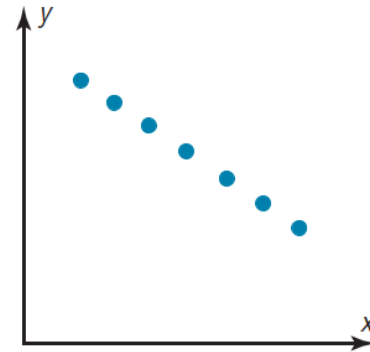
(c) $r = 1.00$



(d) $r = -0.50$



(e) $r = -0.90$



(f) $r = -1.00$



An Introduction to Interpolation



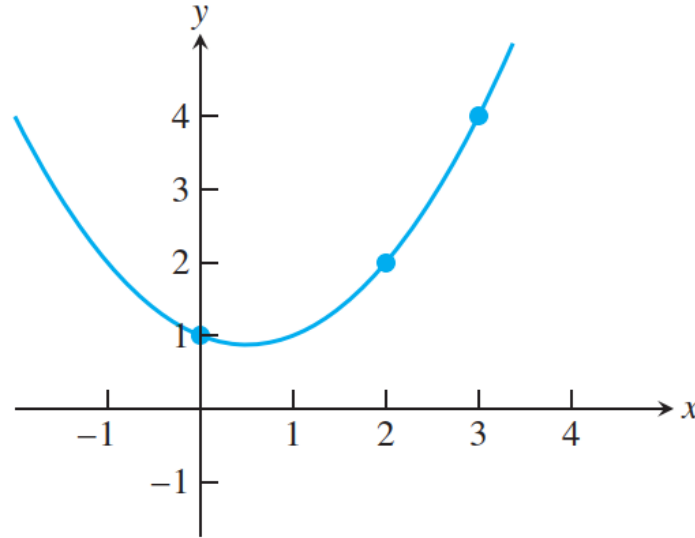
Motivation

Efficient ways of representing data are fundamental to advancing the understanding of scientific problems. At its most fundamental, approximating data by a polynomial is an act of data compression. Suppose that points (x, y) are taken from a given function $y = f(x)$, or perhaps from an experiment where x denotes temperature and y denotes reaction rate. A function on the real numbers represents an infinite amount of information. Finding a polynomial through the set of data means replacing the information with a rule that can be evaluated in a finite number of steps. Although it is unrealistic to expect the polynomial to represent the function exactly at new inputs x , it may be close enough to solve practical problems.



Interpolating functions

A function is said to interpolate a set of data points if it passes through those points. Suppose that a set of (x, y) data points has been collected, such as $(0, 1)$, $(2, 2)$, and $(3, 4)$. There is a parabola that passes through the three points, shown in Figure. This parabola is called the degree 2 interpolating polynomial passing through the three points.



Interpolating functions

The function $y = P(x)$ **interpolates** the data points $(x_1, y_1), \dots, (x_n, y_n)$ if $P(x_i) = y_i$ for each $1 \leq i \leq n$.

- Interpolation is the reverse of evaluation.
- In **polynomial evaluation**, we are given a polynomial and asked to evaluate a y -value for a given x -value—that is, compute points lying on the curve.
- **Polynomial interpolation** asks for the opposite process: Given these points, compute a polynomial that can generate them.



Lagrange interpolation

Assume that n data points $(x_1, y_1), \dots, (x_n, y_n)$ are given, and that we would like to find an interpolating polynomial. There is an explicit formula, called the Lagrange interpolating formula, for writing down a polynomial of degree $d = n - 1$ that interpolates the points. For example, suppose that we are given three points $(x_1, y_1), (x_2, y_2), (x_3, y_3)$. Then the polynomial

$$P_2(x) = y_1 \frac{(x - x_2)(x - x_3)}{(x_1 - x_2)(x_1 - x_3)} + y_2 \frac{(x - x_1)(x - x_3)}{(x_2 - x_1)(x_2 - x_3)} + y_3 \frac{(x - x_1)(x - x_2)}{(x_3 - x_1)(x_3 - x_2)}$$

is the **Lagrange interpolating polynomial** for these points. First notice why the points each lie on the polynomial curve. When x_1 is substituted for x , the terms evaluate to $y_1 + 0 + 0 = y_1$. The second and third numerators are chosen to disappear when x_1 is substituted, and the first denominator is chosen just so to balance the first denominator so that y_1 pops out. It is similar when x_2 and x_3 are substituted.

Main Theorem of Polynomial Interpolation. Let $(x_1, y_1), \dots, (x_n, y_n)$ be n points in the plane with distinct x_i . Then there exists one and only one polynomial P of degree $n - 1$ or less that satisfies $P(x_i) = y_i$ for $i = 1, \dots, n$.

Example 5

Find the polynomial of degree 3 or less that interpolates the points $(0, 2)$, $(1, 1)$, $(2, 0)$, and $(3, -1)$.

SOLUTION The Lagrange form is as follows:

$$\begin{aligned} P(x) &= 2 \frac{(x-1)(x-2)(x-3)}{(0-1)(0-2)(0-3)} + 1 \frac{(x-0)(x-2)(x-3)}{(1-0)(1-2)(1-3)} \\ &\quad + 0 \frac{(x-0)(x-1)(x-3)}{(2-0)(2-1)(2-3)} - 1 \frac{(x-0)(x-1)(x-2)}{(3-0)(3-1)(3-2)} \\ &= -\frac{1}{3}(x^3 - 6x^2 + 11x - 6) + \frac{1}{2}(x^3 - 5x^2 + 6x) - \frac{1}{6}(x^3 - 3x^2 + 2x) \\ &= -x + 2. \end{aligned}$$



An Introduction to Curve Fitting



Inconsistent systems of equations

It is not hard to write down a system of equations that has no solutions. Consider the following three equations in two unknowns:

$$x_1 + x_2 = 2$$

$$x_1 - x_2 = 1$$

$$x_1 + x_2 = 3. \quad (1)$$

Any solution must satisfy the first and third equations, which cannot both be true. A system of equations with no solution is called **inconsistent**.

What is the meaning of a system with no solutions? Perhaps the coefficients are slightly inaccurate. In many cases, the number of equations is greater than the number of unknown variables, making it unlikely that a solution can satisfy all the equations. In fact, m equations in n unknowns typically have no solution when $m > n$. Even though Gaussian elimination will not give us a solution to an inconsistent system $Ax = b$, we should not completely give up. An alternative in this situation is to find a vector x that comes the closest to being a solution.



Inconsistent systems of equations

If we choose this “closeness” to mean close in Euclidean distance, there is a straightforward algorithm for finding the closest x . This special x will be called the **least squares solution**.

We can get a better picture of the failure of system (1) to have a solution by writing it in a different way. The matrix form of the system is $Ax = b$, or

$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}. \quad (2)$$

The alternative view of matrix/vector multiplication is to write the equivalent equation

$$x_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}. \quad (3)$$



Inconsistent systems of equations

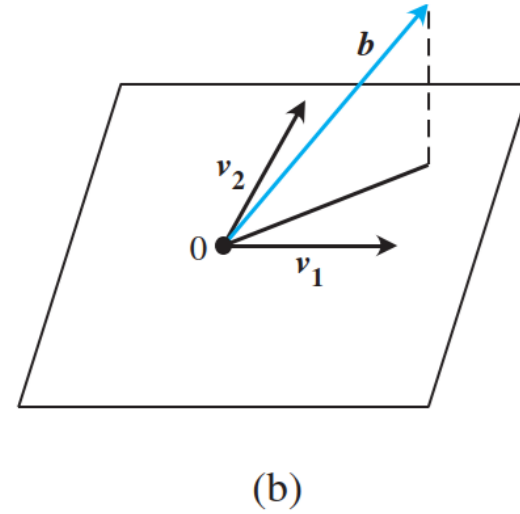
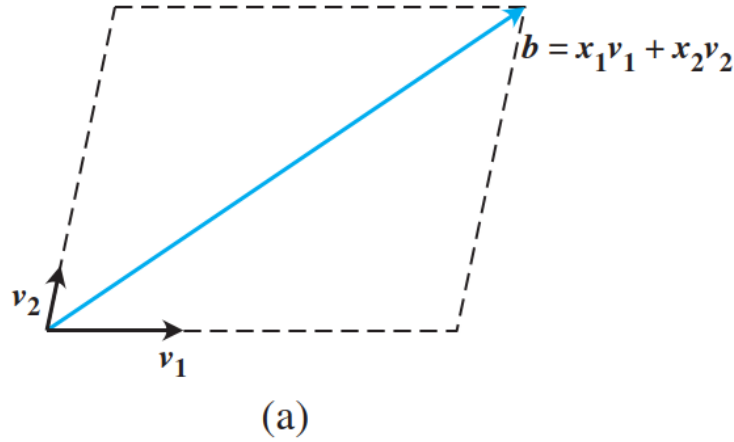
In fact, any $m \times n$ system $Ax = b$ can be viewed as a vector equation

$$x_1 v_1 + x_2 v_2 + \cdots + x_n v_n = b, \quad (4)$$

which expresses b as a linear combination of the columns v_i of A , with coefficients $x_1, \dots, x_n$. In our case, we are trying to hit the target vector b as a linear combination of two other three-dimensional vectors. Since the combinations of two three-dimensional vectors form a plane inside R^3 , equation (3) has a solution only if the vector b lies in that plane. This will always be the situation when we are trying to solve m equations in n unknowns, with $m > n$. Too many equations make the problem overspecified and the equations inconsistent.



Inconsistent systems of equations



Geometric solution of a system of three equations in two unknowns.

(a) Equation (3) requires that the vector b , the right-hand side of the equation, is a linear combination of the columns vectors v_1 and v_2 . (b) If b lies outside of the plane defined by v_1 and v_2 , there will be no solution. The least squares solution $\bar{x}$ makes the combination vector $A\bar{x}$ the one in the plane Ax that is nearest to b in the sense of Euclidean distance.



Inconsistent systems of equations

Figure (b) shows a direction for us to go when a solution does not exist. There is no pair x_1, x_2 that solves $(\quad)$, but there is a point in the plane Ax of all possible candidates that is closest to b . This special vector $A\bar{x}$ is distinguished by the following fact: The residual vector $b - A\bar{x}$ is perpendicular to the plane $\{Ax | x \in R^n\}$. We will exploit this fact to find a formula for $\bar{x}$, the least squares “solution.”

First we establish some notation. Recall the concept of the **transpose** A^T of the $m \times n$ matrix A , which is the $n \times m$ matrix whose rows are the columns of A and whose columns are the rows of A , in the same order. The transpose of the sum of two matrices is the sum of the transposes, $(A + B)^T = A^T + B^T$. The transpose of a product of two matrices is the product of the transposes in the reverse order—that is, $(AB)^T = B^T A^T$.



Inconsistent systems of equations

To work with perpendicularity, recall that two vectors are at right angles to one another if their dot product is zero. For two m -dimensional column vectors u and v , we can write the dot product solely in terms of matrix multiplication by

$$u^T v = [u_1, \dots, u_m] \begin{bmatrix} v_1 \\ \vdots \\ v_m \end{bmatrix}. \quad (5)$$

The vectors u and v are perpendicular, or **orthogonal**, if $u^T \cdot v = 0$, using ordinary matrix multiplication.

Now we return to our search for a formula for $\bar{x}$. We have established that

$$(b - A\bar{x}) \perp \{Ax | x \in R^n\}.$$

Expressing the perpendicularity in terms of matrix multiplication, we find that

$$(Ax)^T (b - A\bar{x}) = 0 \text{ for all } x \text{ in } R^n.$$



Inconsistent systems of equations

Using the preceding fact about transposes, we can rewrite this expression as

$$x^T A^T (b - A\bar{x}) = 0 \text{ for all } x \text{ in } R^n,$$

meaning that the n -dimensional vector $A^T (b - A\bar{x})$ is perpendicular to every vector x in R^n , including itself. There is only one way for that to happen:

$$A^T (b - A\bar{x}) = 0.$$

This gives a system of equations that defines the least squares solution,

$$A^T A\bar{x} = A^T b. \quad (6)$$

The system of equations (6) is known as the **normal equations**. Its solution $\bar{x}$ is the so-called least squares solution of the system $Ax = b$.



Inconsistent systems of equations

Normal equations for least squares

Given the inconsistent system

$$Ax = b,$$

solve

$$A^T A \bar{x} = A^T b$$

for the least squares solution $\bar{x}$ that minimizes the Euclidean length of the residual $r = b - Ax$.



Example 6, revisit slide 29

Use the normal equations to find the least squares solution of the inconsistent system (1).

SOLUTION The problem in matrix form $Ax = b$ has

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}.$$

The components of the normal equations are

$$A^T A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

and

$$A^T b = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}.$$



Example 6

SOLUTION

The normal equations

$$\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$$

can now be solved by Gaussian elimination. The tableau form is

$$\left[\begin{array}{cc|c} 3 & 1 & 6 \\ 1 & 3 & 4 \end{array} \right] \longrightarrow \left[\begin{array}{cc|c} 3 & 1 & 6 \\ 0 & 8/3 & 2 \end{array} \right],$$

which can be solved to get $\bar{x} = (\bar{x}_1, \bar{x}_2) = (7/4, 3/4)$.



Example 6

SOLUTION

Substituting the least squares solution into the original problem yields

$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \frac{7}{4} \\ \frac{3}{4} \end{bmatrix} = \begin{bmatrix} 2.5 \\ 1 \\ 2.5 \end{bmatrix} \neq \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}.$$

To measure our success at fitting the data, we calculate the residual of the least squares solution $\bar{x}$ as

$$r = b - A\bar{x} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} - \begin{bmatrix} 2.5 \\ 1 \\ 2.5 \end{bmatrix} = \begin{bmatrix} -0.5 \\ 0.0 \\ 0.5 \end{bmatrix}.$$

If the residual is the zero vector, then we have solved the original system $Ax = b$ exactly. If not, the Euclidean length of the residual vector is a backward error measure of how far $\bar{x}$ is from being a solution.



Residual quantification

There are at least three ways to express the size of the residual. The Euclidean length of a vector,

$$||r||_2 = \sqrt{r_1^2 + \cdots + r_m^2}, \quad (7)$$

is called the **2-norm**. The **squared error**

$$SE = r_1^2 + \cdots + r_m^2,$$

and the **root mean squared error** (the root of the mean of the squared error)

$$RMSE = \sqrt{SE/m} = \sqrt{(r_1^2 + \cdots + r_m^2)/m}, \quad (8)$$

are also used to measure the error of the least squares solution. The three expressions are closely related; namely

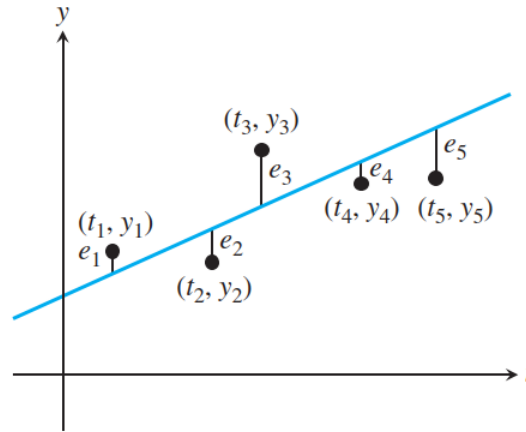
$$RMSE = \frac{\sqrt{SE}}{\sqrt{m}} = \frac{||r||_2}{\sqrt{m}},$$

so finding the $\bar{x}$ that minimizes one, minimizes all. For Example 6, the $SE = (.5)^2 + 0^2 + (-.5)^2 = 0.5$, the 2-norm of the error is $||r||_2 = \sqrt{0.5} \approx 0.707$, and the $RMSE = \sqrt{0.5/3} = 1/\sqrt{6} \approx 0.408$.



Fitting models to data

Let $(t_1, y_1), \dots, (t_m, y_m)$ be a set of points in the plane, which we will often refer to as the “data points.” Given a fixed class of models, such as all lines $y = c_1 + c_2 t$, we can seek to locate the specific instance of the model that best fits the data points in the 2-norm. The core of the least squares idea consists of measuring the residual of the fit by the squared errors of the model at the data points and finding the model parameters that minimize this quantity. This criterion is displayed in Figure.



Fitting models to data

Fitting data by least squares

Given a set of m data points $(t_1, y_1), \dots, (t_m, y_m)$.

STEP 1. Choose a model. Identify a parameterized model, such as $y = c_1 + c_2 t$, which will be used to fit the data.

STEP 2. Force the model to fit the data. Substitute the data points into the model. Each data point creates an equation whose unknowns are the parameters, such as c_1 and c_2 in the line model. This results in a system $Ax = b$, where the unknown x represents the unknown parameters.

STEP 3. Solve the normal equations. The least squares solution for the parameters will be found as the solution to the system of normal equations $A^T A \bar{x} = A^T b$.



Example 7

Find the best line and best parabola for the four data points $(-1, 1)$, $(0, 0)$, $(1, 0)$, $(2, -2)$.

SOLUTION

In accordance with the preceding program, we will follow three steps:

(1) Choose the model $y = c_1 + c_2t$. (2) Forcing the model to fit the data yields

$$c_1 + c_2(-1) = 1$$

$$c_1 + c_2(0) = 0$$

$$c_1 + c_2(1) = 0$$

$$c_1 + c_2(2) = -2,$$

or, in matrix form,

$$\begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -2 \end{bmatrix}.$$



Example 7

SOLUTION

(3) The normal equations are

$$\begin{bmatrix} 4 & 2 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} -1 \\ -5 \end{bmatrix}.$$

Solving for the coefficients c_1 and c_2 results in the best line $y = c_1 + c_2t = 0.2 - 0.9t$.

The residuals are

t	y	line	error
-1	1	1.1	-0.1
0	0	0.2	-0.2
1	0	-0.7	0.7
2	-2	-1.6	-0.4

The error statistics are squared error $SE = (-.1)^2 + (-.2)^2 + (.7)^2 + (-.4)^2 = 0.7$ and $RMSE = \sqrt{.7}/\sqrt{4} = 0.418$.



Example 7

SOLUTION

Next, we extend this example by keeping the same four data points, but changing the model. Set $y = c_1 + c_2t + c_3t^2$ and substitute the data points to yield

$$c_1 + c_2(-1) + c_3(-1)^2 = 1$$

$$c_1 + c_2(0) + c_3(0)^2 = 0$$

$$c_1 + c_2(1) + c_3(1)^2 = 0$$

$$c_1 + c_2(2) + c_3(2)^2 = -2,$$

or, in matrix form,

$$\begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -2 \end{bmatrix}.$$



Example 7

SOLUTION This time, the normal equations are three equations in three unknowns:

$$\begin{bmatrix} 4 & 2 & 6 \\ 2 & 6 & 8 \\ 6 & 8 & 18 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} -1 \\ -5 \\ -7 \end{bmatrix}.$$

Solving for the coefficients results in the best parabola $y = c_1 + c_2t + c_3t^2 = 0.45 - 0.65t - 0.25t^2$. The residual errors are given in the following table:

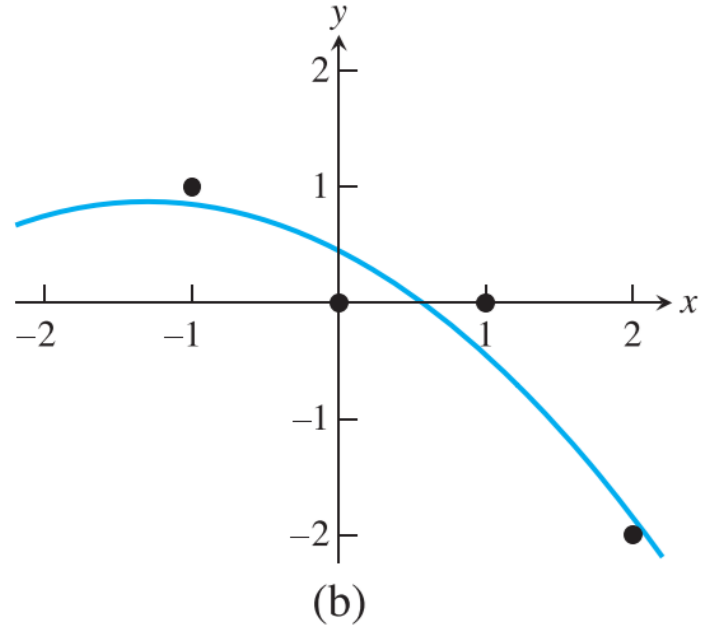
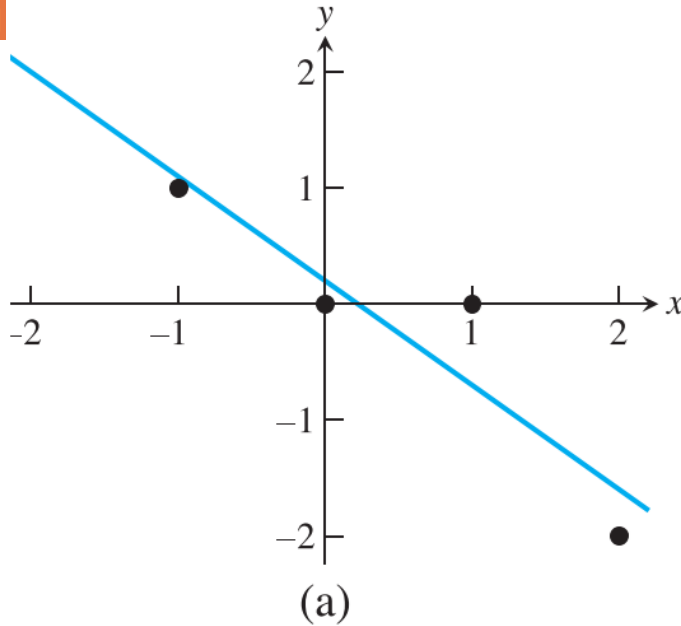
t	y	parabola	error
-1	1	0.85	0.15
0	0	0.45	-0.45
1	0	-0.45	0.45
2	-2	-1.85	-0.15

The error statistics are squared error $SE = (.15)^2 + (-.45)^2 + (.45)^2 + (-.15)^2 = 0.45$ and $RMSE = \sqrt{.45}/\sqrt{4} \approx 0.335$.



Example 7

SOLUTION



Least Squares Fits to Data Points in Example 7 (a) Best line

$y = 0.2 - 0.9t$. RMSE is 0.418. (b) Best parabola $y = 0.45 - 0.65t - 0.25t^2$. RMSE is 0.335.



Periodic data modeling

The previous linear and polynomial models illustrate the use of least squares to fit data. The art of data modeling includes a wide variety of models, some derived from physical principles underlying the source of the data and others based on empirical factors.

Periodic data calls for periodic models. Outside air temperatures, for example, obey cycles on numerous timescales, including daily and yearly cycles governed by the rotation of the earth and the revolution of the earth around the sun.



Example 8

Fit the recorded temperatures in Washington, D.C., on January 1, 2001, as listed in the following table, to a periodic model:

time of day	t	temp (C)
12 mid.	0	-2.2
3 am	$\frac{1}{8}$	-2.8
6 am	$\frac{1}{4}$	-6.1
9 am	$\frac{3}{8}$	-3.9
12 noon	$\frac{1}{2}$	0.0
3 pm	$\frac{5}{8}$	1.1
6 pm	$\frac{3}{4}$	-0.6
9 pm	$\frac{7}{8}$	-1.1

SOLUTION

We choose the model $y = c_1 + c_2 \cos 2\pi t + c_3 \sin 2\pi t$ to match the fact that temperature is roughly periodic with a period of 24 hours, at least in the absence of longer-term temperature movements. The model uses this information by fixing the period to be exactly one day, where we are using days for the t units. The variable t is listed in these units in the table.



Example 8

SOLUTION

Substituting the data into the model results in the following overdetermined system of linear equations:

$$c_1 + c_2 \cos 2\pi(0) + c_3 \sin 2\pi(0) = -2.2$$

$$c_1 + c_2 \cos 2\pi \left(\frac{1}{8} \right) + c_3 \sin 2\pi \left(\frac{1}{8} \right) = -2.8$$

$$c_1 + c_2 \cos 2\pi \left(\frac{1}{4} \right) + c_3 \sin 2\pi \left(\frac{1}{4} \right) = -6.1$$

$$c_1 + c_2 \cos 2\pi \left(\frac{3}{8} \right) + c_3 \sin 2\pi \left(\frac{3}{8} \right) = -3.9$$

$$c_1 + c_2 \cos 2\pi \left(\frac{1}{2} \right) + c_3 \sin 2\pi \left(\frac{1}{2} \right) = 0.0$$

$$c_1 + c_2 \cos 2\pi \left(\frac{5}{8} \right) + c_3 \sin 2\pi \left(\frac{5}{8} \right) = 1.1$$

$$c_1 + c_2 \cos 2\pi \left(\frac{3}{4} \right) + c_3 \sin 2\pi \left(\frac{3}{4} \right) = -0.6$$

$$c_1 + c_2 \cos 2\pi \left(\frac{7}{8} \right) + c_3 \sin 2\pi \left(\frac{7}{8} \right) = -1.1$$



Example 8

SOLUTION

The corresponding inconsistent matrix equation is $Ax = b$, where

$$A = \begin{bmatrix} 1 & \cos 0 & \sin 0 \\ 1 & \cos \frac{\pi}{4} & \sin \frac{\pi}{4} \\ 1 & \cos \frac{\pi}{2} & \sin \frac{\pi}{2} \\ 1 & \cos \frac{3\pi}{4} & \sin \frac{3\pi}{4} \\ 1 & \cos \pi & \sin \pi \\ 1 & \cos \frac{5\pi}{4} & \sin \frac{5\pi}{4} \\ 1 & \cos \frac{3\pi}{2} & \sin \frac{3\pi}{2} \\ 1 & \cos \frac{7\pi}{4} & \sin \frac{7\pi}{4} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & \sqrt{2}/2 & \sqrt{2}/2 \\ 1 & 0 & 1 \\ 1 & -\sqrt{2}/2 & \sqrt{2}/2 \\ 1 & -1 & 0 \\ 1 & -\sqrt{2}/2 & -\sqrt{2}/2 \\ 1 & 0 & -1 \\ 1 & \sqrt{2}/2 & -\sqrt{2}/2 \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} -2.2 \\ -2.8 \\ -6.1 \\ -3.9 \\ 0.0 \\ 1.1 \\ -0.6 \\ -1.1 \end{bmatrix}.$$

The normal equations $A^T A c = A^T b$ are

$$\begin{bmatrix} 8 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} -15.6 \\ -2.9778 \\ -10.2376 \end{bmatrix},$$

which are easily solved as $c_1 = -1.95$, $c_2 = -0.7445$, and $c_3 = -2.5594$. The best version of the model, in the sense of least squares, is $y = -1.9500 - 0.7445 \cos 2\pi t - 2.5594 \sin 2\pi t$, with $\text{RMSE} \approx 1.063$.



Example 9, revisit slide 49

Fit the temperature data to the improved model

$$y = c_1 + c_2 \cos 2\pi t + c_3 \sin 2\pi t + c_4 \cos 4\pi t. \quad (9)$$

SOLUTION The system of equations is now

$$c_1 + c_2 \cos 2\pi(0) + c_3 \sin 2\pi(0) + c_4 \cos 4\pi(0) = -2.2$$

$$c_1 + c_2 \cos 2\pi\left(\frac{1}{8}\right) + c_3 \sin 2\pi\left(\frac{1}{8}\right) + c_4 \cos 4\pi\left(\frac{1}{8}\right) = -2.8$$

$$c_1 + c_2 \cos 2\pi\left(\frac{1}{4}\right) + c_3 \sin 2\pi\left(\frac{1}{4}\right) + c_4 \cos 4\pi\left(\frac{1}{4}\right) = -6.1$$

$$c_1 + c_2 \cos 2\pi\left(\frac{3}{8}\right) + c_3 \sin 2\pi\left(\frac{3}{8}\right) + c_4 \cos 4\pi\left(\frac{3}{8}\right) = -3.9$$

$$c_1 + c_2 \cos 2\pi\left(\frac{1}{2}\right) + c_3 \sin 2\pi\left(\frac{1}{2}\right) + c_4 \cos 4\pi\left(\frac{1}{2}\right) = 0.0$$

$$c_1 + c_2 \cos 2\pi\left(\frac{5}{8}\right) + c_3 \sin 2\pi\left(\frac{5}{8}\right) + c_4 \cos 4\pi\left(\frac{5}{8}\right) = 1.1$$

$$c_1 + c_2 \cos 2\pi\left(\frac{3}{4}\right) + c_3 \sin 2\pi\left(\frac{3}{4}\right) + c_4 \cos 4\pi\left(\frac{3}{4}\right) = -0.6$$

$$c_1 + c_2 \cos 2\pi\left(\frac{7}{8}\right) + c_3 \sin 2\pi\left(\frac{7}{8}\right) + c_4 \cos 4\pi\left(\frac{7}{8}\right) = -1.1,$$



Example 9

SOLUTION

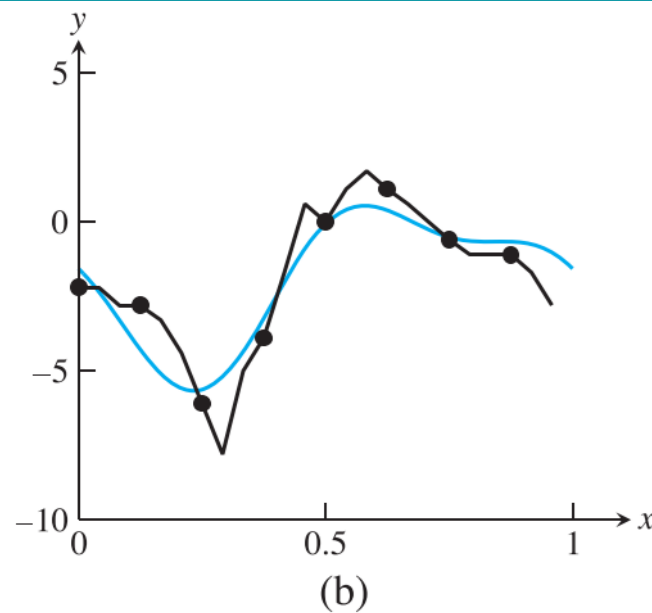
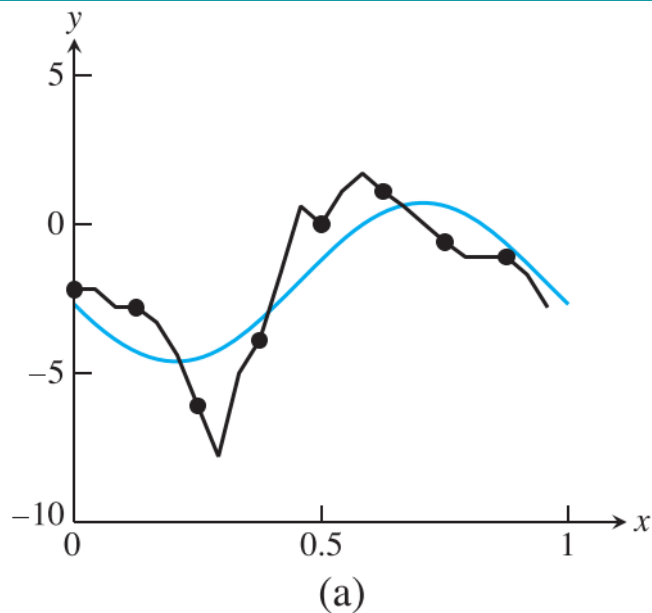
leading to the following normal equations:

$$\begin{bmatrix} 8 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix} = \begin{bmatrix} -15.6 \\ -2.9778 \\ -10.2376 \\ 4.5 \end{bmatrix}.$$

The solutions are $c_1 = -1.95$, $c_2 = -0.7445$, $c_3 = -2.5594$, and $c_4 = 1.125$, with $\text{RMSE} \approx 0.705$.



Benchmarking the periodic models in examples 8, 9 for the given data



Least Squares Fits to Periodic Data in Examples (a) **8**: Sinusoid model $y = -1.95 - 0.7445 \cos 2\pi t - 2.5594 \sin 2\pi t$ shown in bold, along with recorded temperature trace on Jan 1, 2001. (b) **9**: Improved sinusoid $y = -1.95 - 0.7445 \cos 2\pi t - 2.5594 \sin 2\pi t + 1.125 \cos 4\pi t$ fits the data more closely.



Data linearization: The exponential model

Exponential growth of a population is implied when its rate of change is proportional to its size. Under perfect conditions, when the growth environment is unchanging and when the population is well below the carrying capacity of the environment, the model is a good representation.

The **exponential model**

$$y = c_1 e^{c_2 t} \quad (10)$$

cannot be directly fit by least squares because c_2 does not appear linearly in the model equation.



Data linearization: The exponential model

In the case of the exponential model (10), the model is linearized by applying the natural logarithm:

$$\ln y = \ln(c_1 e^{c_2 t}) = \ln c_1 + c_2 t. \quad (11)$$

Note that for an exponential model, the graph of $\ln y$ is a linear plot in t . At first glance, it appears that we have only traded one problem for another. The c_2 coefficient is now linear in the model, but c_1 no longer is. However, by renaming $k = \ln c_1$, we can write

$$\ln y = k + c_2 t. \quad (12)$$

Now both coefficients k and c_2 are linear in the model. After solving the normal equations for the best k and c_2 , we can find the corresponding $c_1 = e^k$ if we wish.



Example 10

The number of transistors on Intel central processing units since the early 1970s is given in the table that follows. Fit the model $y = c_1 e^{c_2 t}$ to the data.

CPU	year	transistors
4004	1971	2,250
8008	1972	2,500
8080	1974	5,000
8086	1978	29,000
286	1982	120,000
386	1985	275,000
486	1989	1,180,000
Pentium	1993	3,100,000
Pentium II	1997	7,500,000
Pentium III	1999	24,000,000
Pentium 4	2000	42,000,000
Itanium	2002	220,000,000
Itanium 2	2003	410,000,000



Example 10

SOLUTION

Parameters will be fit by using model linearization (11). Linearizing the model gives

$$\ln y = k + c_2 t.$$

We will let $t = 0$ correspond to the year 1970. Substituting the data into the linearized model yields

$$k + c_2(1) = \ln 2250$$

$$k + c_2(2) = \ln 2500$$

$$k + c_2(4) = \ln 5000$$

$$k + c_2(8) = \ln 29000,$$

and so forth. The matrix equation is $Ax = b$, where $x = (k, c_2)$,



Example 10

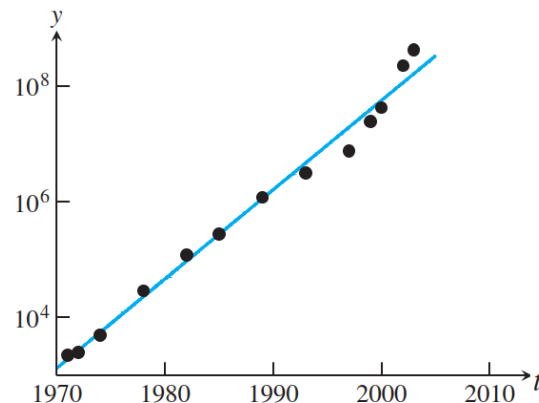
SOLUTION

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 4 \\ 1 & 8 \\ \vdots & \vdots \\ 1 & 33 \end{bmatrix}, \text{ and } b = \begin{bmatrix} \ln 2250 \\ \ln 2500 \\ \ln 5000 \\ \ln 29000 \\ \vdots \\ \ln 410000000 \end{bmatrix}.$$

The normal equations $A^T A \bar{x} = A^T b$ are

$$\begin{bmatrix} 13 & 235 \\ 235 & 5927 \end{bmatrix} \begin{bmatrix} k \\ c_2 \end{bmatrix} = \begin{bmatrix} 176.90 \\ 3793.23 \end{bmatrix},$$

which has solution $k \approx 7.197$ and $c_2 \approx 0.3546$, leading to $c_1 = e^k \approx 1335.3$. The exponential curve $y = 1335.3e^{0.3546t}$ is shown in Figure along with the data. The doubling time for the law is $\ln 2/c_2 \approx 1.95$ years. Gordon C. Moore, cofounder of Intel, predicted in 1965 that over the ensuing decade, computing power would double every 2 years. Astoundingly, that exponential rate has continued for 40 years. There is some evidence in Figure that this rate has accelerated since 2000.



Semilog Plot of Moore's Law. Number of transistors on CPU chip versus year.



Data linearization: The power law

Another important example with nonlinear coefficients is the **power law** model $y = c_1 t^{c_2}$. This model also can be simplified with linearization by taking logs of both sides:

$$\begin{aligned}\ln y &= \ln c_1 + c_2 \ln t \\ &= k + c_2 \ln t.\end{aligned}$$

Substitution of data into the model will give

$$\begin{aligned}k + c_2 \ln t_1 &= \ln y_1 \\ &\vdots \\ k + c_2 \ln t_n &= \ln y_n,\end{aligned}$$

resulting in the matrix form

$$A = \begin{bmatrix} 1 & \ln t_1 \\ \vdots & \vdots \\ 1 & \ln t_n \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} \ln y_1 \\ \vdots \\ \ln y_n \end{bmatrix}.$$

The normal equations allow determination of k and c_2 , and $c_1 = e^k$.



Example 11

Use linearization to fit the given height–weight data with a power law model.

The mean height and weight of boys ages 2–11 were collected in the U.S. National Health and Nutrition Examination Survey by the Centers for Disease Control (CDC) in 2002, resulting in the following table:

age (yrs.)	height (m)	weight (kg)
2	0.9120	13.7
3	0.9860	15.9
4	1.0600	18.5
5	1.1300	21.3
6	1.1900	23.5
7	1.2600	27.2
8	1.3200	32.7
9	1.3800	36.0
10	1.4100	38.6
11	1.4900	43.7



Example 11

SOLUTION

Following the preceding strategy, the resulting power law for weight versus height is $W = 16.3H^{2.42}$. The relationship is graphed in Figure. Since weight is a proxy for volume, the coefficient $c_2 \approx 2.42$ can be viewed as the “effective dimension” of the human body.

